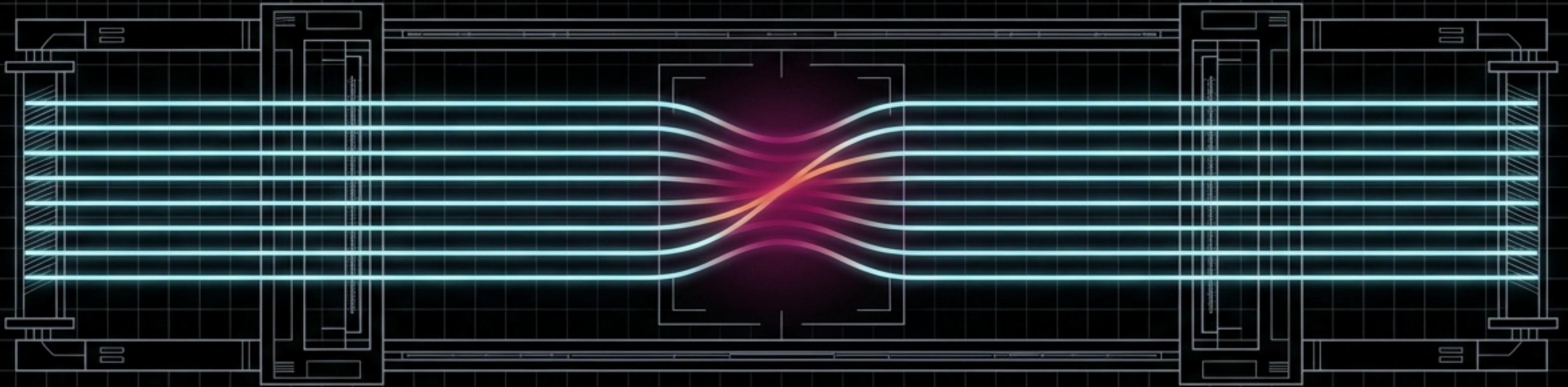


Q-Weave: Crosstalk-Aware Error Mitigation

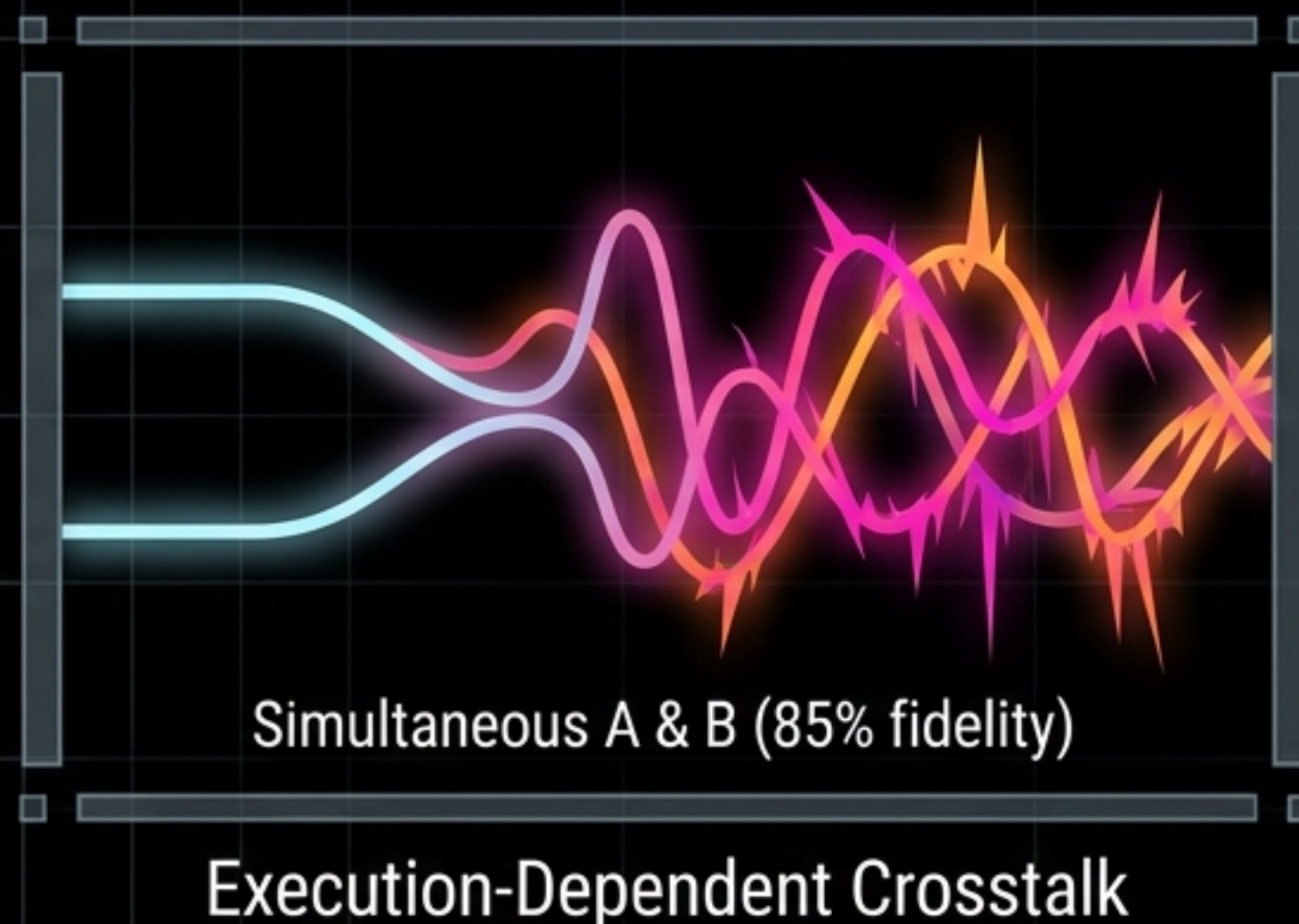


An Empirical Framework for Crosstalk-Aware Error
Mitigation in NISQ Quantum Circuits.

Parallel execution is an illusion in NISQ processors.



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The Problem

Isolated gate-error information cannot predict the quality of an entire circuit execution.

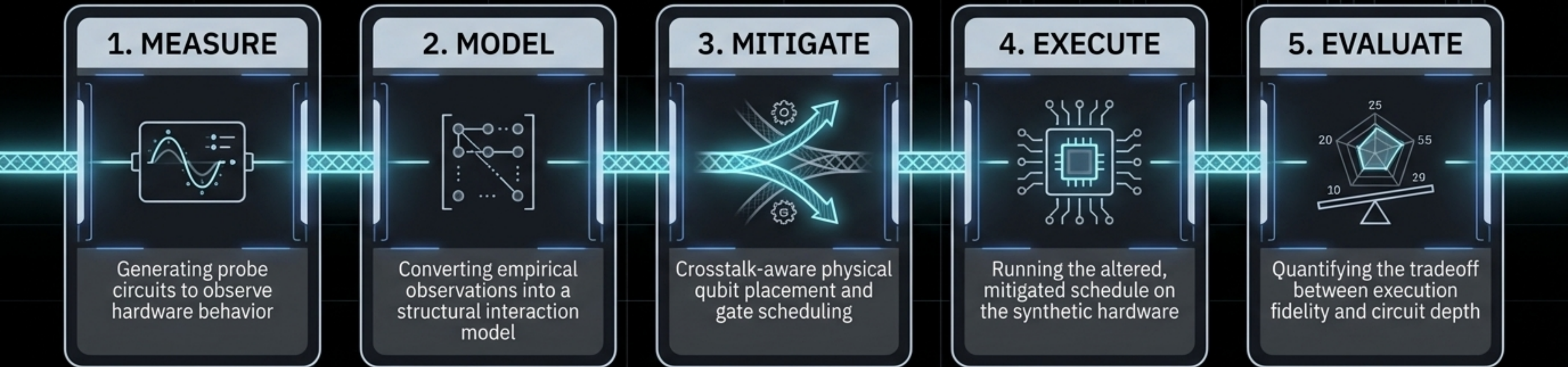
The Trap

A circuit schedule that maximizes parallelism to reduce decoherence time may actually destroy fidelity because simultaneous operations physically interfere.

Defining the boundaries of the Q-Weave framework.

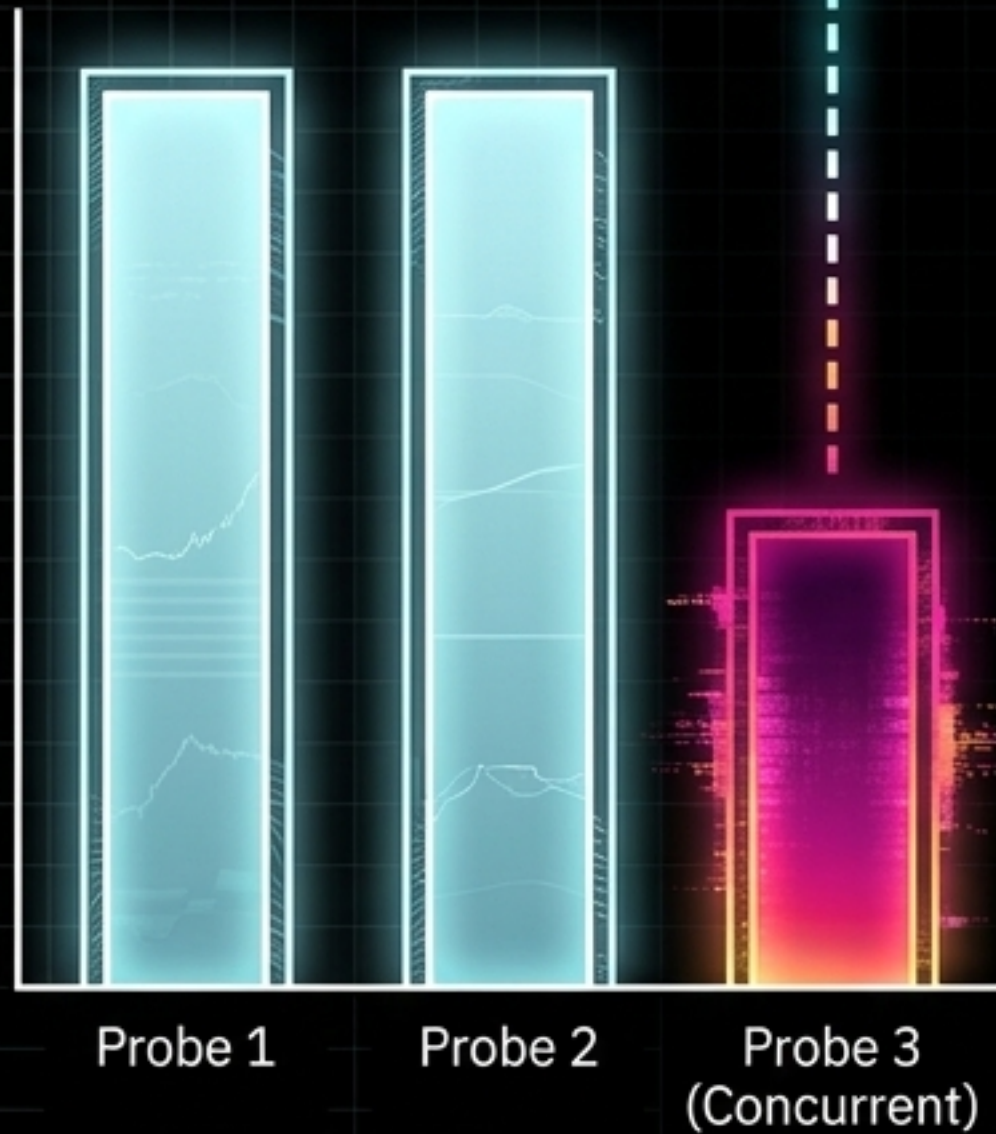
Domain:	<u>Quantum Error Correction</u> (Logical Qubits)	Software-Level Error Mitigation (Compilation tweaks).
Scope:	<u>Just a routing/scheduling</u> algorithm	A Full Empirical Pipeline (Measure → Model → Mitigate).
Method:	<u>AI / Machine Learning</u> optimization	Empirical Measurement & Greedy Heuristics.
Platform:	A Qiskit extension / plugin	An Independent Framework (Using Qiskit merely as an experimental testbed).

The lifecycle of a crosstalk-aware quantum circuit

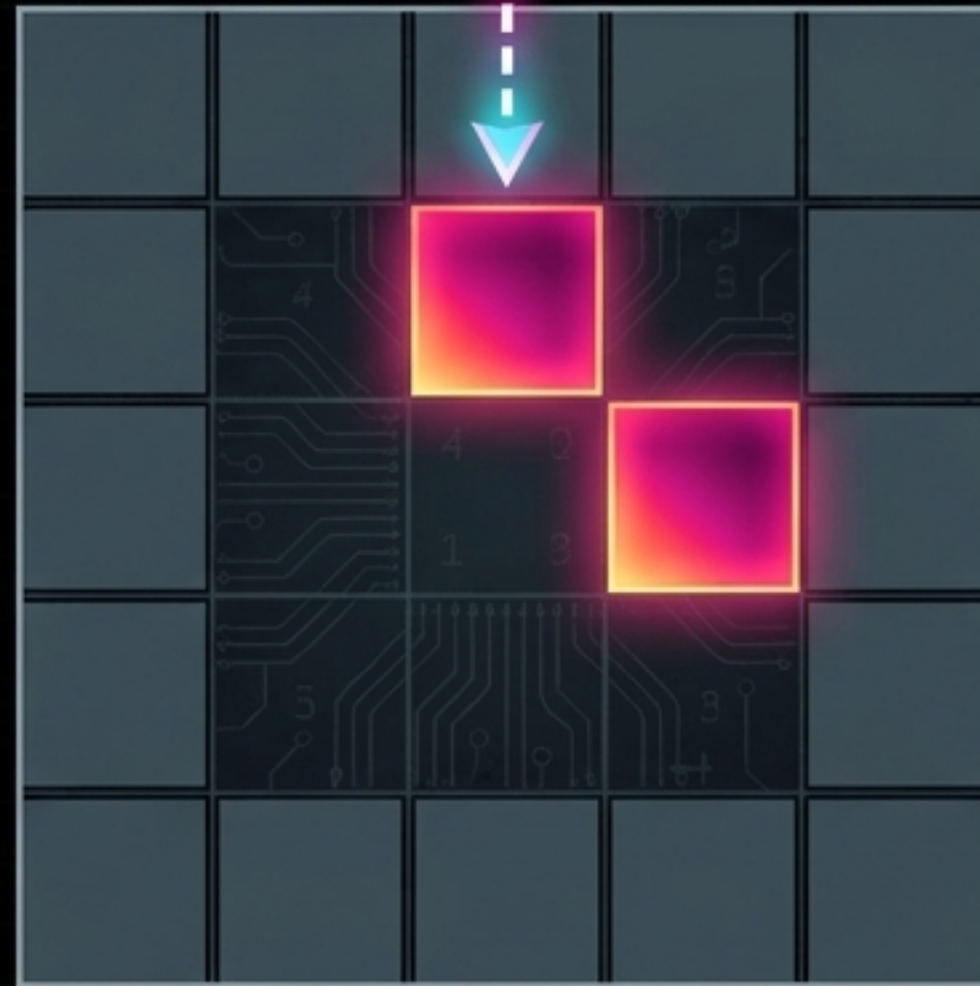


Translating physical interference into compiler logic.

Measurement



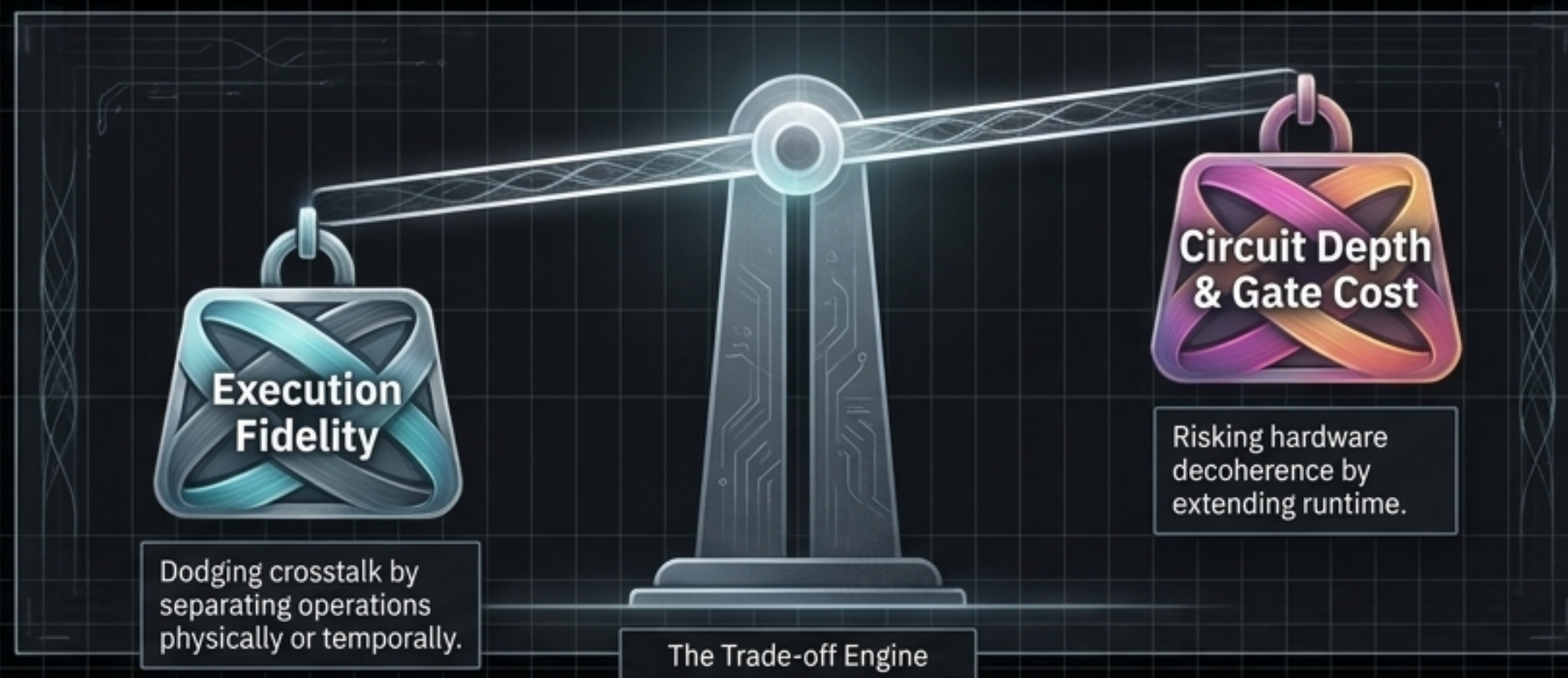
Interaction Matrix



Hardware Graph

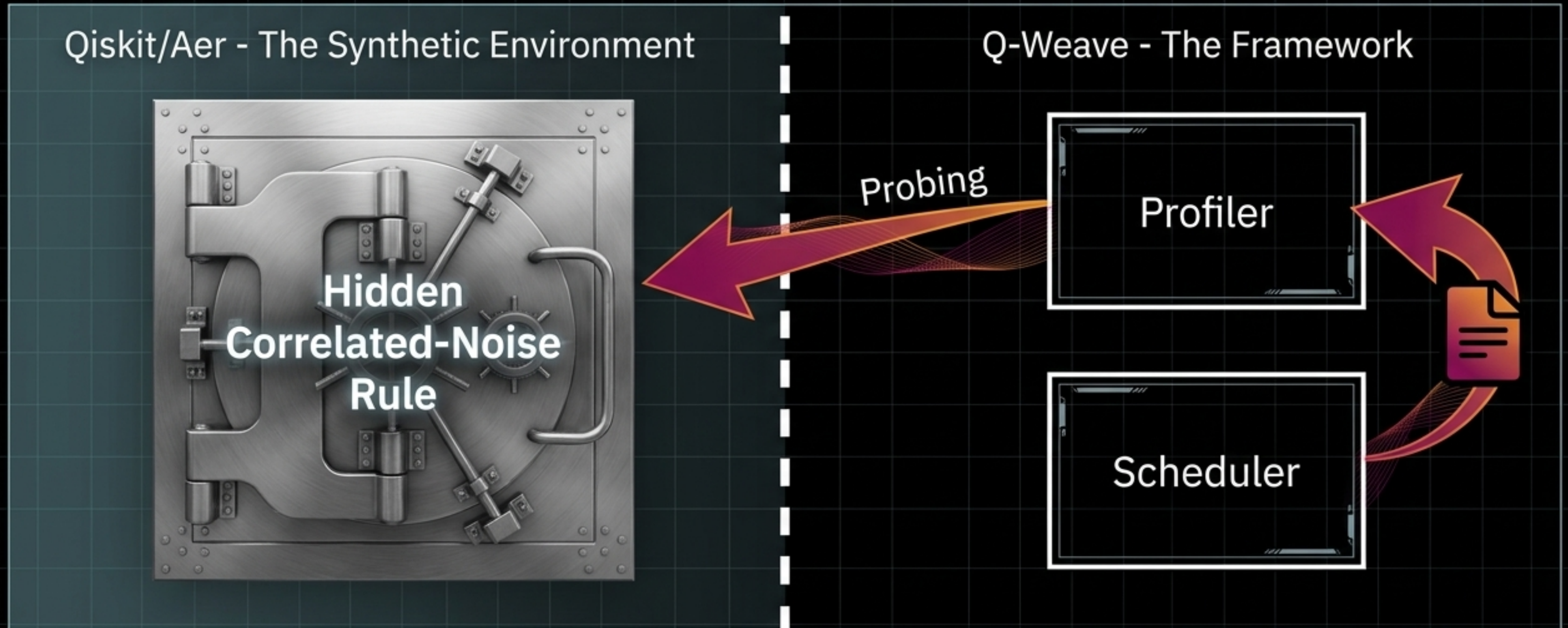


Balancing fidelity gains against circuit depth and decoherence.



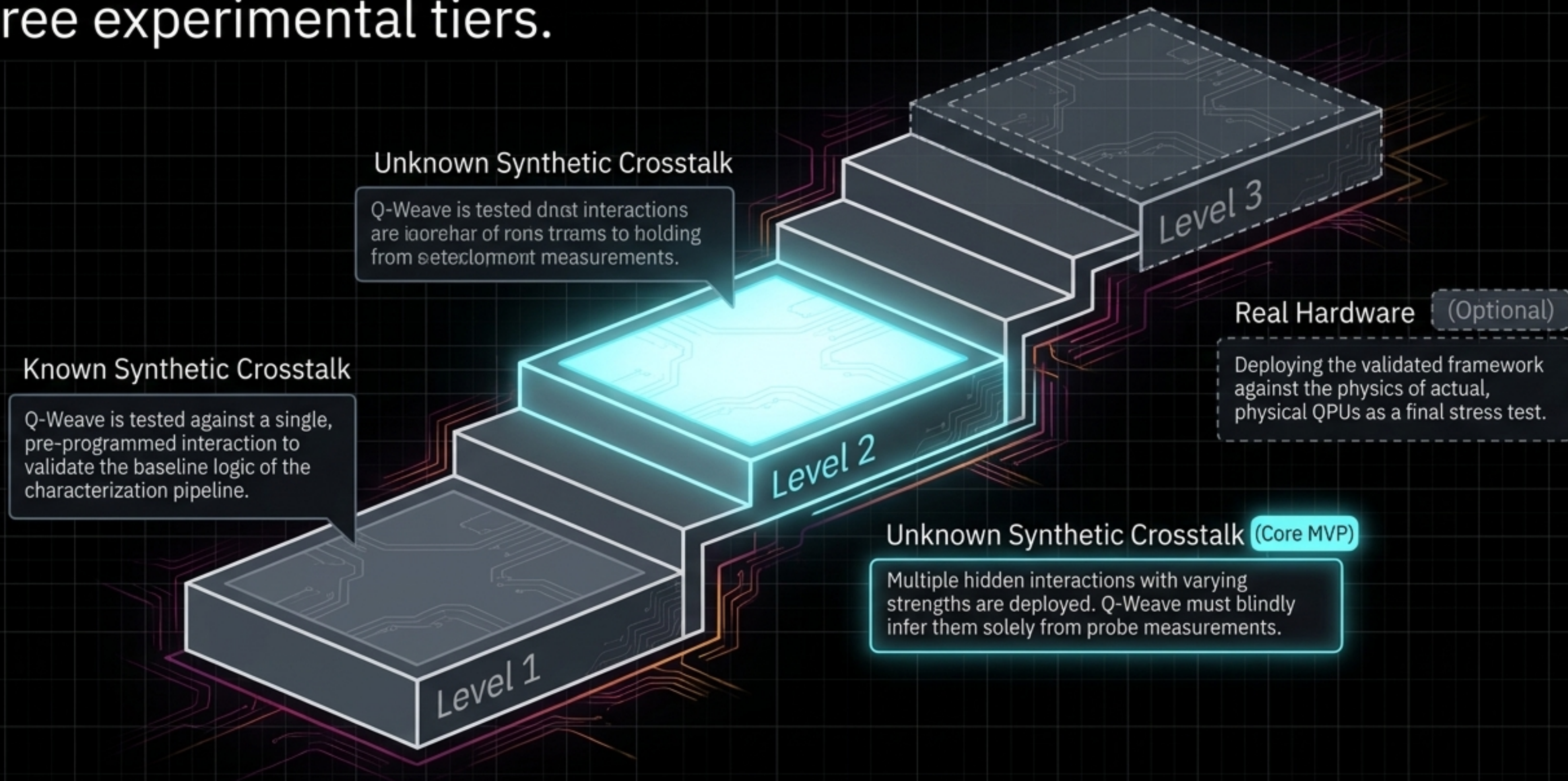
Placement	Scheduling (Greedy Heuristic)
Avoiding mapping heavily interacting logical operations into regions with high measured $W(e)$.	Q-Weave does not minimize crosstalk at any cost. It serializes interacting gates just enough to find a low-cost equilibrium, increasing depth strictly to dodge the most destructive friction.

Isolating the empirical framework from the synthetic environment



The profiler is strictly prohibited from reading the simulator's hardware logic. Q-Weave must empirically discover crosstalk from raw measurements, proving the framework's real-world viability.

Validating the pipeline across three experimental tiers.



Success is measured by tradeoff quantification, not absolute percentages.



1. Detection

Successfully identifying hidden synthetic rules from probe measurements alone.



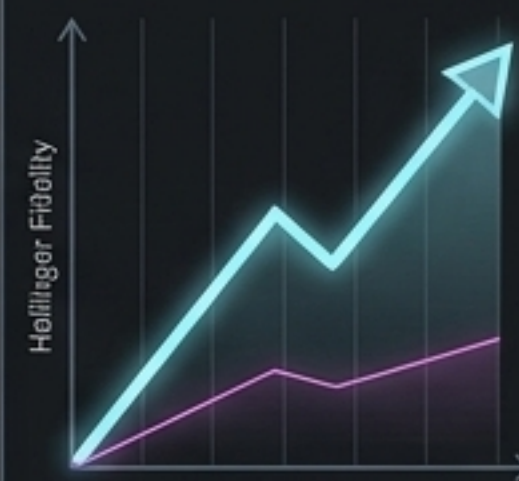
2. Modeling

Accurately mapping those findings into a usable hardware graph.



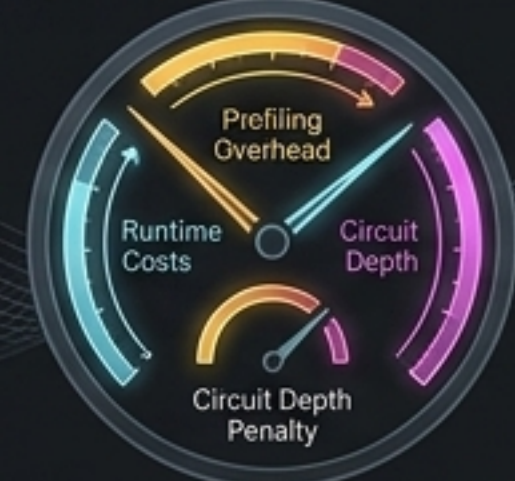
3. Mitigation

The compiler demonstrably altering execution schedules based on the graph.



4. Improvement

Achieving higher expected Hellinger fidelity compared to the isolated baseline.



5. Cost Measurement

Quantifying profiling overhead, runtime costs, and the exact circuit depth penalty incurred.

The empirical framework for NISQ error mitigation.

Q-Weave empirically discovers execution-dependent crosstalk, models those interactions as a weighted hardware graph, and uses the model for software-level error mitigation through crosstalk-aware qubit placement and scheduling, with Qiskit/Aer used as the experimental platform.